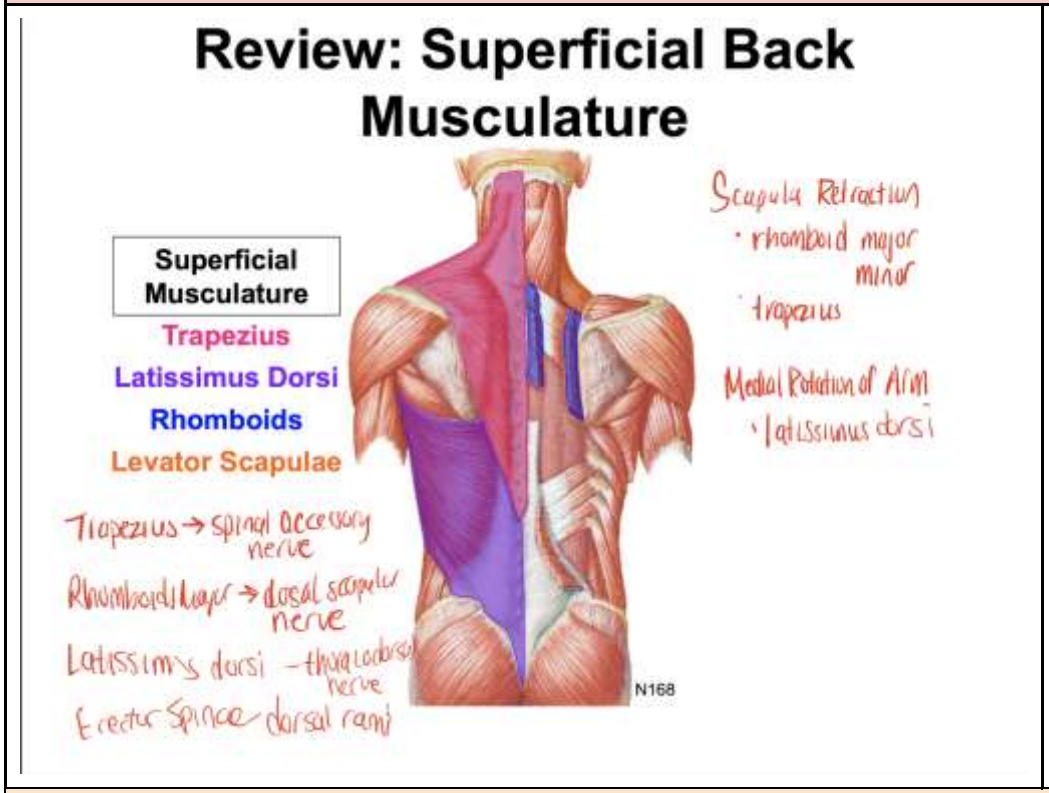
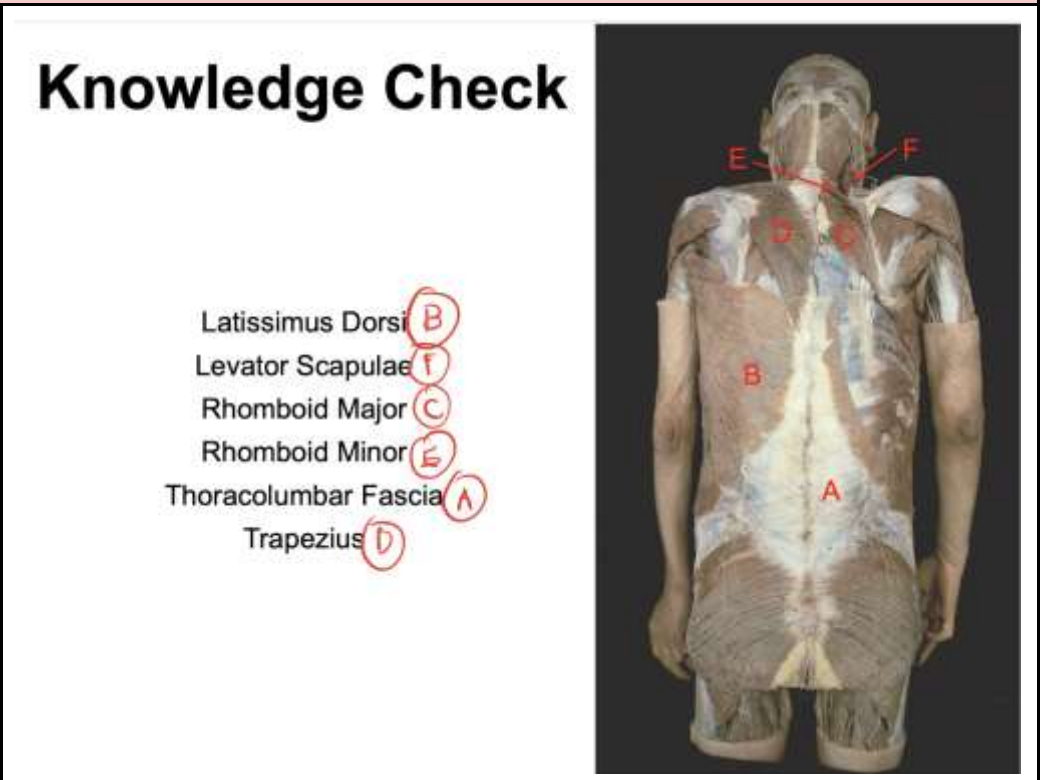
Week #8: [Muscles of the Shoulder]

Review: Muscles of the Superficial Back
Muscles of the Pectoral Region
Muscles of the Shoulder
Approach to Rotator Cuff Disease **(Dr. Faber)**

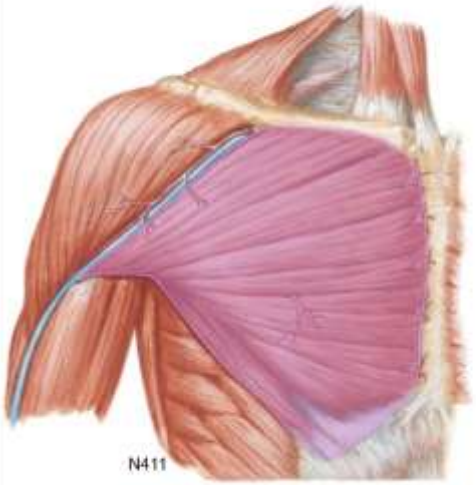
Review: Muscles of the Superficial Back

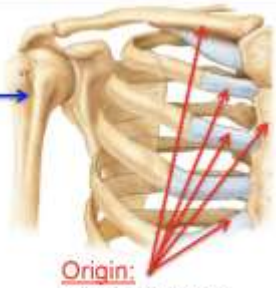




Muscles of the Pectoral Region

Pectoralis Major



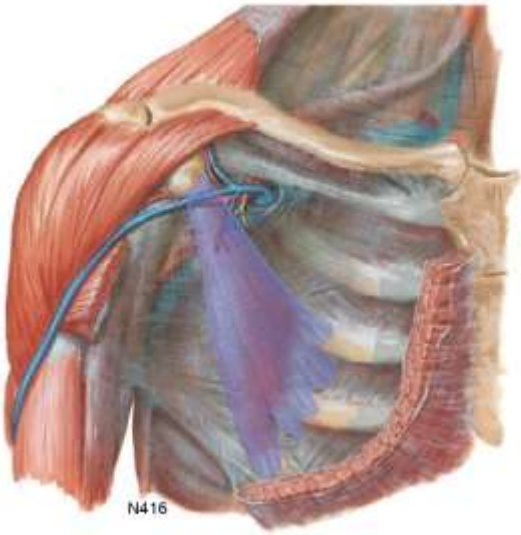


Insertion:
Lateral lip of intertubercular groove

Origin:
Medial Clavicle
Sternum
Upper 6 Ribs and Costal Cartilages

- Function:
  - Adducts, flexes & medially rotates humerus
- Innervation:
  - Medial & Lateral Pectoral Nerves

Pectoralis Minor



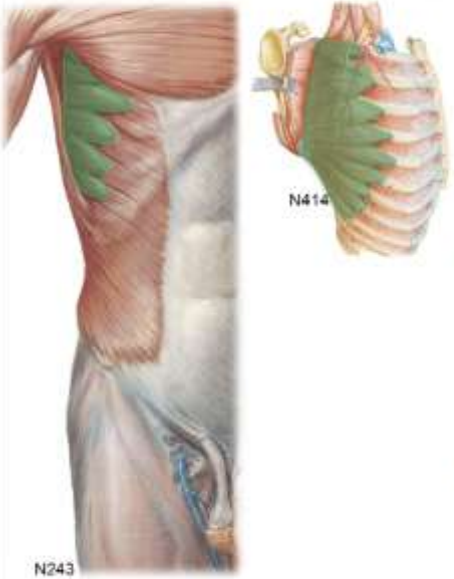


Insertion:
Coracoid Process

Origin:
Ribs 3-5

- Function:
  - Stabilizes scapula by drawing it inferiorly and anteriorly against thoracic wall
- Innervation:
  - Medial & Lateral Pectoral Nerves

Serratus Anterior





Insertion:
Medial Border of Scapula

Origin:
Ribs 1-8

- Function:
  - Protraction of the scapula
- Innervation:
  - Long Thoracic Nerve (C5,6,7)

Muscle	Origin	Insertion	Innervation	Function
Pectoralis Major	Medial Clavicle Lateral Sternum Upper 6 ribs & costal cartilages	Anterior lateral lip of the intertubercular groove/sulcus	Medial and Lateral Pectoral Nerves	Adducts, flexes, and medially rotates humerus
Pectoralis Minor	Ribs 3-5 (near costal cartilages)	Coracoid Process		Stabilizes scapula by drawing it anteriorly and inferiorly against thoracic wall
Serratus Anterior	Ribs 1-8	Medial Border of Scapula	Long Thoracic Nerve	Protraction of scapula

Clinical Note: If the nerve supply to the serratus anterior is damaged. The serratus anterior is no longer able to hold the scapula against the posterior thoracic wall while pushing with the upper extremity. This results in a **winged scapula presentation.**

Serratus Anterior: Winged Scapula

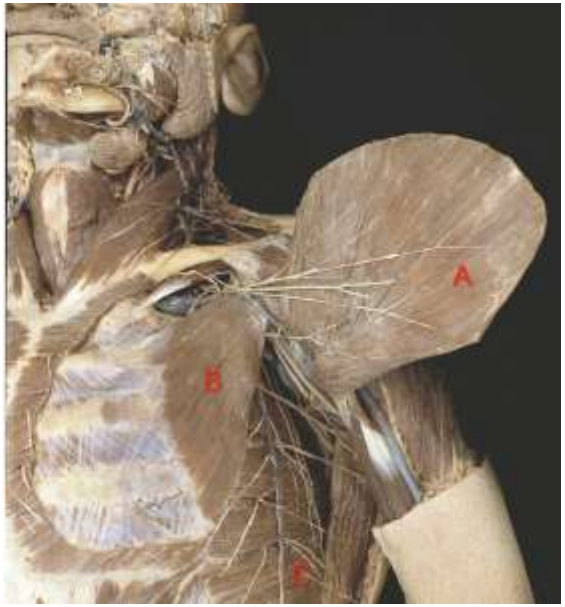
Long Thoracic Nerve Lesion:

- Will lead to ***winging of the scapula***
  - the serratus anterior is unable to keep the medial border of the scapula against the thoracic wall when pushing with the upper limb



Knowledge Check

Draws Scapula Inferiorly (B)
Flexes the Arm (A)
Protracts the Scapula (C)



Muscles of the Shoulder

Deltoid



N411



Origin:
Lateral 1/3 of clavicle
Acromion
Spine of scapula



Insertion:
Deltoid tuberosity

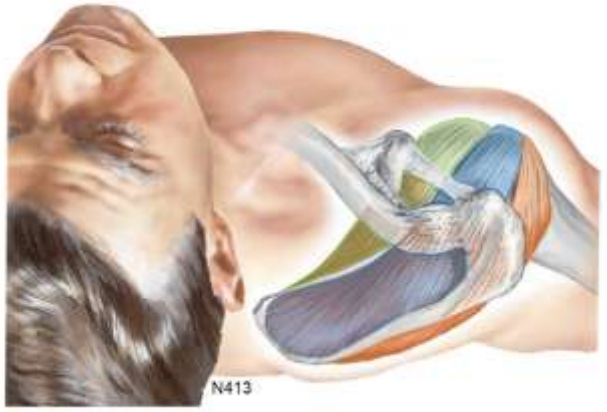
- Function:
  - Clavicular/Anterior Head: flexes & medially rotates arm
  - Acromial/Middle Head: abducts arm
  - Spinal/Posterior Head: extends & laterally rotates arm
- Innervation:
  - Axillary Nerve (C5, C6)



N410

Rotator Cuff

Supraspinatus
Infraspinatus
Subscapularis
Teres Minor



N413



Anterior



Posterior

Supraspinatus



Anterior





Origin:
Supraspinous Fossa

Insertion:
Greater Tubercle



Posterior

- Function:
  - First 15° of arm abduction
  - Stabilizes glenohumeral joint
- Innervation:
  - Suprascapular Nerve (C5, 6)



Infraspinatus



Anterior



Origin:
Infraspinous Fossa

Insertion:
Greater Tubercle

- Function:
  - Lateral rotation of arm
  - Stabilizes glenohumeral joint



Posterior

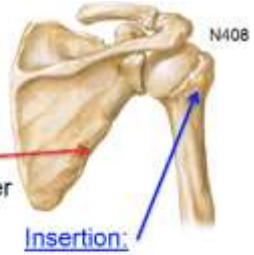
- Innervation:
  - Suprascapular Nerve (C5, 6)



Teres Minor



Anterior



Origin:
Lateral border of scapula

Insertion:
Greater Tubercle

- Function:
  - Lateral rotation of arm
  - Stabilizes glenohumeral joint



Posterior

- Innervation:
  - Axillary Nerve (C5, 6)



Subscapularis



Anterior



Insertion:
Lesser Tubercle

Origin:
Subscapular Fossa

- Function:
  - Medially rotates arm
  - Stabilizes glenohumeral joint



Posterior

- Innervation:
  - Upper & Lower Subscapular Nerves (C5, 6, 7)



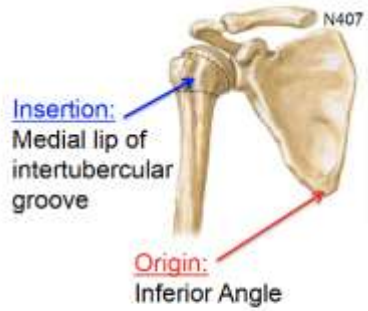
Teres Major



Anterior



Posterior

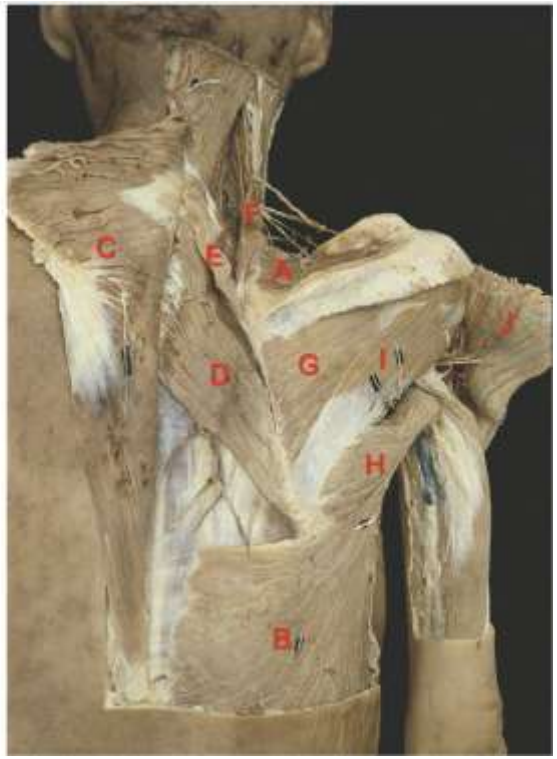


- Function:
  - Adducts and medially rotates arm
- Innervation:
  - Lower Subscapular Nerve (C5,6)

Supposed to be ADduct

Knowledge Check

Deltoid J
Infraspinatus G
Levator Scapula F
Latissimus Dorsi B
Rhomboid Major D
Rhomboid Minor E
Supraspinatus A
Teres Major H
Teres Minor I
Trapezius C



Muscles of the Shoulder Region

Muscle	Origin	Insertion	Innervation	Function
Deltoid	Lateral 1/3 Clavicle, Acromion, Spine of Scapula	Deltoid Tuberosity	Axillary Nerve	Ant: Flexes and medially rotates arm Mid: Abducts humerus Post: Extends and laterally rotates arm
Teres Major	Inferior Angle of Scapula	Medial lip of intertubercular groove	Lower Subscapular Nerve	Abducts and medially rotates humerus
Supraspinatus	Supraspinous Fossa	Greater Tubercle	Suprascapular Nerve	Abduction (first 15 degrees) Stabilizes GH joint
Infraspinatus	Infraspinous Fossa	Greater Tubercle	Suprascapular Nerve	Lateral rotation of arm Stabilizes GH joint
Teres Minor	Lateral border of scapula	Greater Tubercle	Axillary Nerve	Lateral rotation of arm Stabilizes GH joint
Subscapularis	Subscapular Fossa	Lesser Tubercle	Upper and Lower Subscapular Nerves	Medial rotation of arm Stabilizes GH joint

ADDuct

Approach to Rotator Cuff Disease (Dr. Faber)